

Anesth-on-the-Go: Designing Portable Game-based Anesthetic Simulator for Education

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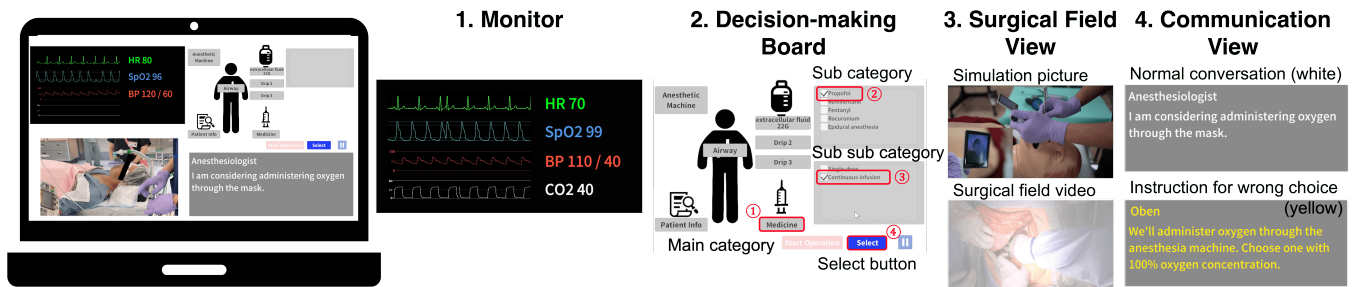


Figure 1: Overview of Anesth-on-the-Go system interface. Interface consists of 4 components; 1) Monitor, 2) Decision-making Board, 3) Surgical Field View, 4) Communication view. The user selects the appropriate procedure on the decision-making board based on the vital signs on the monitor, the surgical field, and the conversations among the surgeons, and proceeds with the surgery.

ABSTRACT

Gaining extensive practice in anesthesia procedures is critical for medical students. However, the current high-fidelity simulators for clinical anesthesia practice limit such vital opportunities due to their professional-level functions, large size, and high installation costs. In this paper, we propose Anesth-on-the-Go, a portable game-based anesthesia simulator that allows medical students to repeatedly practice clinical anesthesia procedures anywhere, anytime on a conventional personal computer. The proposed simulator is designed for medical students, and it allows them to manipulate controls appropriately for various anesthesia procedures according to pre-determined surgical scenarios. Based on iterative interviews with clinical anesthesia instructors, we designed a simulation interface with four key components: 1) a monitor, 2) a decision-making board, 3) a surgical field view, and 4) a communication view. As our initial evaluation, we introduced the simulator in the training curriculum for medical students and verified its effect on their proficiency level in subsequent training with high-fidelity simulation. The results show that individual and prior training with our simulator improved each student's overall performance (i.e., score) and

stimulated discussion among teammates during the subsequent formal training. The evaluation's findings also indicate that the simulator experience can facilitate post-training reflection.

CCS CONCEPTS

• Human-centered computing → Graphical user interfaces.

KEYWORDS

Anesthesiology, surgical simulation, graphical interface

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1 INTRODUCTION

Clinical training and practice in the administration of anesthesia is an essential part of the curriculum for medical students. While repetitive training is essential to master the relevant procedures and knowledge, there are still limited opportunities for such training available to medical students. One explanation for this educational bottleneck is the simulation system needed for training. In most cases, an anesthetic training curriculum implements high-fidelity simulations that rely on a variety of sensors designed to simulate biological reactions. Moreover, the kind of simulation machine used is a professional-level model, and thus not readily available

for extensive training. Accordingly, these installation difficulties have considerably limited the training opportunities for medical students.

Therefore, this paper proposes Anesth-on-the-Go, a portable game-based anesthesia simulator that allows students to practice (play) anywhere and anytime, supporting their repetitive and on-demand training needs for learning anesthesia procedures. Previous screen-based anesthesia simulators have been developed in the medical field [6], our simulator is specifically designed as an interactive educational tool for medical students. A unique feature of anesthesia is the requirement to make serious decisions depending on the patient's situation and the surgical procedure. Consequently, the goal of our simulator is to allow medical students to experience a complete set of necessary anesthesia management tasks during a realistic surgical workflow. To achieve a fully portable experience, we focus on an on-screen simulator without any AR/VR devices.

Anesth-on-the-Go runs on standard personal computers or tablets that support mouse or touch input. Figure 1 illustrates the features of our interface; 1) an intraoperative monitor, 2) a decision-making board, 3) a surgical field view, and 4) a team communication view. Once the user selects the scenario, the simulation begins with the virtual entry into the operating room and ends with the awakening of the patient. Through the simulation, trainees attempt to select appropriate medicines and procedures from graphical buttons or menus on the decision-making board. The simulator has an obex system that corrects procedures through the text navigation if the user makes a critical mistake. We first implemented two basic scenarios of laparotomy simulating the surgical workflow according to the medical faculty curriculum. The history of the choices made is automatically recorded and calculated as an overall score for later review.

As an initial evaluation of the simulator, we introduced it in the training curriculum for fifth-year medical students and verified its educational effect on the subsequent training with actual medical devices. We found that using our simulator before the actual training (1) improved each trainee's overall performance (i.e., score), (2) stimulated discussion among teammates during the training, and (3) facilitated post-training reflection (e.g., finding the differences between decisions made on the simulator and on the device).

The main contributions of this work are (1) providing and discussing suitable UI designs for a portable game-based anesthesia simulator through iterative interviews with medical professors and (2) showing early results demonstrating that our simulator offers a valuable educational tool for medical students to enrich their subsequent real-world clinical training in anesthesia.

2 RELATED WORK

Repetitive practice is necessary for medical students to make appropriate decisions throughout the anesthesia process. Therefore, researchers have proposed tools to assist both trainees and instructors. The work most similar to our simulator is the Anesthesia Simulator-Recorder [6] published in 1990, a pioneering software program that simulates patient anesthesia equipment, vital condition monitors, and management options that are operated and displayed on a computer screen, allowing trainees to experience the anesthesia process with multiple critical incidents. Our work

is along the lines of that earlier work, but with two major updates. First, our software is designed for medical students rather than residents [6]. For example, we have added an obex feature that automatically provides quick guidance, especially when students are making serious misjudgments. The second is a complete user interface update with more sophisticated graphical information (e.g., surgical field video and a GUI with a button-based selection board) and modern medical trends. These updates significantly enhance the earlier work, providing a more realistic and engaging experience for today's medical students.

The use of VR/AR technology, in recent years, has become another mainstream approach to developing an anesthesia training system due to its ability to realistically reproduce the surgical environment and support repetitive practice [2]. VR simulations can closely resemble an actual patient's body, reflecting individual anatomy and different anatomical regions. To enhance this reality, researchers have proposed an ultrasound-guided regional anesthesia (UGRA) technique that reproduces the sensation of inserting a needle into the patient's skin [1]. These systems produce simulations that are closer to reality. On the other hand, systems that support decision-making in anesthesia procedures have also been proposed, for example, one study presented a system that uses AR technology to improve awareness of information by displaying augmented vital signs and other information [5]. Other tools have been designed to support decision-making during anesthesia training using mannequins [3]. However, these systems are basically tools that support conventional mannequin-based training scenarios, so they do not support portable individual learning.

Consequently, through our review of existing systems, we were unable to find any affordable software simulator that supports portable and individualized on-demand anesthesia training.

3 ANESTH-ON-THE-GO: ANESTHETIC SIMULATION FOR EDUCATION

3.1 Design Consideration

In this study, we emphasize the support of novice users like medical students in learning general anesthesia. To achieve this goal, we explored the design requirements of the simulation system. At first, we asked anesthesiologists to transcribe the entire flow of administrative anesthesia procedures and put them together in a tree structure for typical scenarios to clarify the overall picture of administering anesthesia. Through months of discussions with anesthesiologists, we attempt to identify key points that should be applied as the simulator features and interface components.

Firstly, an anesthesiologist should always be able to judge the surgical situation and make decisions on appropriate administrations according to the progress of surgery and the patient's condition. Thus, the user is required to be trained to select the appropriate procedure from among all of the possible choices at the training session. Therefore, we focus on facilitating decision-making so that the user can manage the appropriate administration of anesthesia.

Secondly, General anesthesia is mainly examined based on the patient's vital signs and the surgical field's conditions. The vital signs displayed on the monitor, such as heart rate, pulse oximeter, blood pressure, capnography, and temperature, are all very important and enable the anesthesiologists to comprehend the patient's

current status. The system also should provide the monitor's data and the surgical field's visual information.

Additionally, the main purpose of our system is to educate beginning medical students. We employed a facilitator to correct the students' judgments when they failed to correctly manage a procedure, which meets the third design requirement.

Another important aspect of surgery is that the surgeons, anesthesiologists, and nurses all cooperate. To allow the students to experience an actual surgical atmosphere, the monitor displays the conversational communications of the surgeons, anesthesiologists, and nurses.

Finally, we initially employ typical surgery scenarios as the educational material of beginning students. Since the purpose of this education is to learn the entire flow of the anesthetic procedures, we only adopt basic scenarios that adhere to the medical school curriculum.

Based on these features, we eventually specified the following five points as our system design requirements;

- (1) Facilitate decision-making for the general administration of anesthesia.
- (2) Provide monitor and surgical field information.
- (3) Employ a facilitator to correct the students' wrong judgments.
- (4) Show the conversational communication among all of the surgeons, anesthesiologists, and nurses.
- (5) Employ typical surgical scenarios.

3.2 System Implementation

To make the anesthetic training more practical, we built a portable anesthesia simulation system. Following the above design requirements, system implementation proceeded based on two main steps: 1) categorizing all possible options in anesthetic procedures to facilitate decision-making and 2) organizing the complex anesthetic procedures into a system programmable list format.

Although we observed the various anesthetic procedure options through the anesthesiologist's transcribed anesthetic procedure mentioned in section 3.2, these were very disparate. Thus, we organized and classified all of the anesthetic procedure options into five categories: anesthetic machine, airway, medicine, patient information, and drip. We call these five categories "main category" (Appendix Table 1). Then, we applied these categories to our system interface.

Since the textualized surgical scenario was very complicated, we reorganized it into a system-programmable list form. We listed the pairs of current/next steps and the appropriate procedure in Excel format. To apply these anesthetic procedure data to the system, the system converts the Excel format to JSON format, compares the correct answer for each step with the answer for the procedure selected by the user, and then proceeds to the next step if the answer is correct. During the simulation, the system reads the JSON data line by line.

3.3 System Interface

We implemented the anesthetic simulation as a desktop system that operates using the standard interface of a PC monitor and mouse input, which allows the user to practice anytime and anywhere. This

system was made with Unity 2019.03. Figure 1 shows the system interface, and the following explains the details of each part.

1) Monitor (Figure 1-1)

The monitor displays four key vital signs with pulse waves: heart rate, pulse oximeter, blood pressure, and capnography, which provide key information for understanding the patient's condition.

2) Decision-Making Board (Figure 1-2)

There are five main category buttons for anesthetic procedures: anesthetic machine, airway, medicine, drip, and patient information. All five main category buttons and each detail procedure are always displayed, which allows users to select the appropriate one in a similar way to the real surgery environment. When each button is selected, detailed procedure options are shown hierarchically (Sub-category/Sub-sub-category in Appendix Table 1). Users can select appropriate procedures by clicking checkboxes from the board and then proceed to the next step with the selection button.

3) Surgical Field View (Figure 1-3)

This is a normal high-fidelity simulation image or a real patient's abdomen image, which is shown according to the current step. The user refers to this picture or video to decide appropriate actions.

4) Communication View (Figure 1-4)

This area shows the conversations or briefings among all of the surgeons, anesthesiologists, and nurses. Since the operation proceeds with the cooperation of these professionals as a team, verbal communication is also an essential element. Users can observe other's comments as guidance to managing procedures. If the users make a critical mistake, a facilitator called oben suggests the correct procedure and leads them back to the correct approach.

4 APPLICATION FLOW

Here, we show the entire application scenario flow and our simulator interface's behavior. We divide the entire scenario into the five typical phases below.

1) Scenario Selection (Figure 2-1)

We prepare two typical scenarios that are different only in being with/without the bleeding step. Users can check the scenario (patient) details, select a scenario, and then proceed to start the operation.

2) Induction of Anesthesia (Figure 2-2)

Before starting the incision, induction of anesthesia is performed to put the patient to sleep. After selecting and administering an anesthetic from the drug button to put the patient to sleep, the trachea is intubated from the airway button, the airway is secured, and respiration is managed by artificial respiration. The patient's vital signs can be observed on a monitor, which is continuously updated to present the heart rate, pulse oximeter, blood pressure, capnography, and individual pulse waves. Analgesics might also need to be administered in advance, depending on the patient's condition, to prepare for surgical invasion.

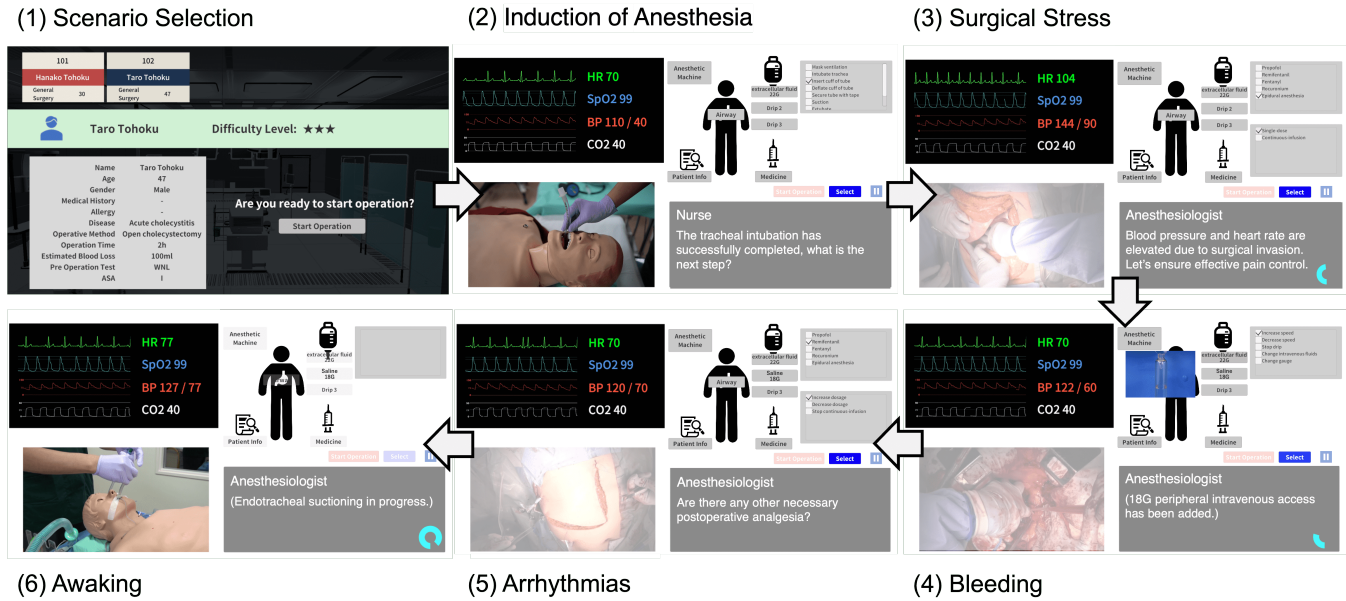


Figure 2: Application scenario timeline

3) Surgical Stress (Figure 2-3)

Here, the surgeon opens the patient's abdomen. The anesthesiologist monitors a significant increase in blood pressure and manages it by administering the appropriate medicine.

4) Bleeding (Figure 2-4)

If bleeding from the affected site occurs during the surgical invasion, appropriate intravenous drugs are administered from the intravenous drip button according to the amount of bleeding. The three infusion buttons are animated when the cursor is moved over them, allowing the user to check the drip rate.

5) Arrhythmias (Figure 2-5)

After the treatment of the affected area is completed, surgeons perform peritoneal lavage. During peritoneal lavage, temporary arrhythmia may occur due to touching the organs. The cause of this arrhythmia must be correctly identified through observation of vital signs and the surgical field to confirm that the arrhythmia was caused by touching the abdomen and that no special medicine are necessary.

6) Awakening (Figure 2-6)

The drug administration is terminated and the patient is awakened from anesthesia. After awakening, the patient's pain status is monitored and analgesics are administered as needed.

5 PRELIMINARY USER STUDY

We conducted a preliminary observational study to determine the effectiveness of our simulator by introducing it in actual anesthesia training for fifth-year medical school students at a local university hospital. We collected feedback about the effect of our simulator on the student's behavior during subsequent clinical practice

from anesthesiologists who regularly supervise the anesthetic training of medical students. We compared the 25 student groups who played/did not play our proposed simulator before actual medical practice. The Department of Anesthesiology organizes compulsory training courses for all fifth-year medical school students and accepts one group per week in turns. Each group had three or four students (fewer than one female student per group on average) participating in the training session as a team. For both conditions, we randomly selected 25 groups from this training. The university's ethics review board approved the study.

5.1 Procedure

Training sessions with our proposed system were introduced before medical practice during a five-day period of anesthetic training. Only one group with 3-4 students attended per training term. On the second day's morning, students learned general anesthesia, and then we provided them with our simulator and asked them to play it on the same day in the afternoon. Students played the typical general anesthetic scenario without the bleeding phase, as explained in Section 4. Each group member collaboratively played the scenario and tried to complete the mission (Figure 3a). The average task completion time was about 20 minutes.

After practicing the anesthetic procedure with our simulator, they conducted regular clinical practice on the last day of the training term with high-fidelity simulations (Figure 3b) at the accreditation center (Figure 3c). The instructor observed the students' training during the clinical practice session and scored their behaviors and anesthetic procedures. The adequate procedure has been graded with 1-2 points for every important step that the instructor has defined beforehand (Appendix Table 2). Following the pre-defined evaluation score, the same instructor graded and calculated the overall score for all 50 groups. We objectively evaluated the

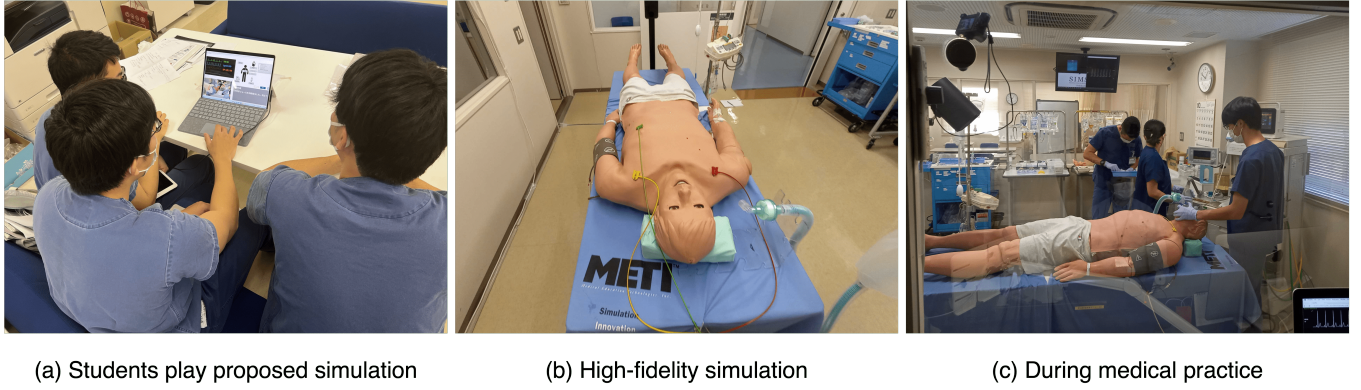


Figure 3: (a) Student group plays proposed simulator before clinical practice. (b) Scene of clinical practice for fifth-year medical students at the accreditation center of a local hospital. (c) High-fidelity simulation for surgical training.

overall scores of clinical training. We also conducted a post-study interview with instructors about how students' behaviors changed after introducing our proposed simulation before clinical practice and how useful the proposed simulation was for the educational curriculum.

5.2 Results

5.2.1 Objective Evaluation. We compared the overall scores for the anesthetic procedure during clinical practice between the groups who played the proposed simulation (2023) and students who did not play the simulation but studied two years ago (2021) under the same scoring protocol. To make such a between-subject comparison for non-parametric datasets, we performed a Mann-Whitney U test on the total scores. The results are shown in Figure 4. We also found that there was a significant difference between the two datasets, demonstrating that the with-simulator condition produced higher scores than the without-simulator condition ($p < .05, Z = 2.54$). The maximum training score was 100.

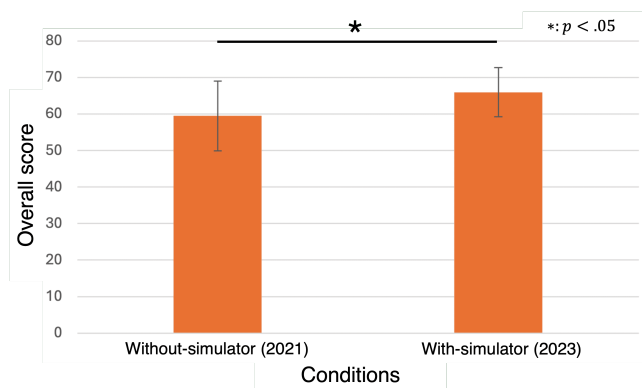


Figure 4: Results of overall training scores: without-simulator condition (2021) vs. with-simulator (Anes-on-the-Go) condition (2023)

This between-subjects comparison was not completely fair because the participants in the with-simulator condition (2023) repeatedly experienced similar training and decision-making processes. On the other hand, the without-simulator participants (2021) had to deal with events during surgery without prior knowledge. However, our primary goal is finding ways to enrich the clinical training of anesthesia, which is generally short with limited opportunities to practice the related decision-making tasks. Toward this aim, the proposed method of deploying our simulator at the beginning of training could be a reasonable solution and should be highly recommended.

5.2.2 Anesthesiologist feedback. The anesthesiology instructor commented that our proposed system could promote the students' active learning, enabling students to manipulate devices themselves and thus enhancing the current commercial video-based materials. The students actively discussed issues with each other based on their prior simulator experiences during the clinical practice. They gave comments such as "We made this choice at this point using the simulation." From such feedback, students seemed to apply the prior experience and knowledge from the simulator to their clinical practice. From the teacher's perspective, the simulator system made it possible to review the students' behaviors more smoothly right after clinical practice corresponding to the simulation's scenario. On the other hand, students often made wrong choices, although they had experienced the same surgical scenario in clinical practice and simulation. This observation suggests that such behavioral training with the physical simulator is still essential, while simulation-based learning is effective for acquiring general knowledge of anesthesia [4]. Furthermore, the instructor felt that it's valuable for students to easily fail without consequence, which naturally is unacceptable in real surgery on a human body.

6 DISCUSSION

UI design considerations. Our four-view design worked well throughout the studies in terms of processing all of the necessary information at once. On the other hand, we observed a clear difference from actual training and surgery from a user experience

perspective. Our current 2D interface does not apply the spatial configuration among the equipment, monitors, patients, team members, and surgeons in an actual operating room. The VR-based approach would be promising to strengthen spatial factors. The reproduced 3D operating room allows users to manage any anesthesia machine, change drip packs, administrate medicine with a syringe, and communicate with other surgeons in as close to the real environment as possible. While various future work is formulated, in the nature of the medical training setup, we reconfirmed that such a classic four-split view interface is still recommended because it allows the learners to focus on the primary process away from advanced interactions.

Deployability. Due to the Anes-on-the-Go system's portability, it has already been used for several years in the local university's medical training programs. Although our first simulator is made with Unity and run on Windows-based devices, Web-based systems would enhance its deployability. The video and design of the current version is the result of years of experience. We have received very positive feedback from anesthesia faculty from a practical perspective, in terms of not only the learning effect of students but also in improving teaching of the instructor. Duplicating the instructor's knowledge into automatic clinical training simulation would reduce the instructors' burden because they currently teach and grade by manpower. We will continue to use this simulator in practice and report more comprehensive findings in the future.

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A APPENDIX

A.1 Full list of anesthetic options

Table 1: All anesthetic options

Main Category	Sub-category	Sub-sub-category
Anesthetic machine	Air 6L/min	
	Air 3L/min	
	Oxygen 6L/min	
	Oxygen 3L/min	
	Air 2L/min + Oxygen 1L/min	
Airway	Mask ventilation	
	Intubate trachea	
	Insert cuff of the tube	
	Deflate the cuff of the tube	
	Secure the tube with tape	
	Suction	
	Extubate	
	Auscultate	
Medicine	Oxygen administration	
	Propofol	Single-dose
	Remifentanyl	Continuous-infusion
	Fentanyl	Stop continuous-infusion
	Rocuronium	Increase dosage
	Sugammadex	Decrease dosage
	Ephedrine	
	Phenylephrine	
	Nicorandil	
	Nicardipine	
	Landiolol	
	Acetaminophen	
	Epidural anesthesia	
Drip	Saline	22G
	Extracellular fluid	20G
	Colloid solution	18G
	Blood transfusion	Increase speed
		Decrease speed
		End drip
		Change intravenous fluids
Patient Information		Change gauge
	Blood gas analysis	
	Surgery progress	
	Bleeding amount	
	Surgical field visualization	

A.2 Evaluation criteria for user study

Table 2: Excerpt of simulation training evaluation index

Evaluation question	Correct procedure	Score
Introduction of anesthesia		
(1) What procedure do they apply for anesthesia induction?	Preoxygenation: administrate 100% oxygen Administration of analgesic drug and sedative drug Confirm sleeping Administrate muscle relaxant Mask ventilation	1 point each
(2) Is the anesthesia induction order appropriate?	Confirm mask ventilation validity	1 point
(3) Do they apply mask ventilation?	Appropriate order	1 point
(4) What analgesics do they administrate?	Did properly Epidural administrate of local anesthetic Administrate Fentanyl Increase Remifentanyl	1 point (any choice)
(5) What procedure do they apply for tracheal intubation?	Take a sniffing position Insert blade Insert tube at appropriate depth Put cuff Secure the tube with tape Auscultation	1 point each
(6) Is the tracheal intubation order appropriate?	After check of chest movement and EtCO ₂ value	1 point
(7) Has SpO ₂ percentage decreased during anesthetic induction?	Appropriate order	2 points
⋮	No changed	
⋮		
Intraoperative management		
(8) Do they administrate analgesics after increased blood pressure?	Increase Remifentanyl Remifentanyl bolus Administrate Fentanyl Increase Fentanyl Increase Sevoflurane	1 point (any choice)
⋮		
⋮		
Bleeding management		
(9) What procedure do they apply for tracheal intubation?	Accelerated infusion speed Accelerated infusion speed more Colloid infusion Vascular access Vascular access with more than 20-gauge intravenous catheter Did blood gas analysis 2nd blood gas analysis Use vasopressor Use Phenylephrine Continue Phenylephrine	1 point each
⋮		
⋮		
Abdominal closure and after surgery care		
(10) Do they notice arrhythmia during abdominal irrigation?	noticed	1 point
(11) What procedure do they apply for postoperative analgesia?	Administrate Fentanyl IV Single-dose epidural anesthesia Increase continuous Fentanyl Flush continuous Fentanyl Apply PCA with epidural anesthesia Start continuous Fentanyl Start continuous epidural anesthesia Already administrated continuous adjuvant analgesics	2 points (any choice)
(12) Do they start continuous analgesic after patient complains of pain?	Already administrated continuous analgesic Start continuous analgesic after complaining	2 points (any procedure)